

HOMework

The DO...WHILE Loop

Pseudocode practice — the repetitive structure with a final test

Solve each exercise below using pseudocode. Use only the do...while structure (do ... while <condition>) — combined with read, write, and simple expressions — to solve each problem. Remember: in a do...while loop the instructions always run at least once, before the condition is ever checked.

Reminder — syntax: `do <instructions> while <condition>`

Exercise 1 — Countdown

LEVEL 1 • BEGINNER

Read a positive integer n . Using a do...while structure, display the numbers from n down to 1, one at a time.

Exercise 2 — Reverse a Number

LEVEL 2 • EASY

Read a positive integer n . Using do...while, display its digits in reverse order. For example, if $n = 1234$, the algorithm should display 4321.

Hint: extract the last digit with $n \% 10$, then remove it with $n \leftarrow [n / 10]$ — this is exactly the pattern used to count digits, but this time you build a new number instead of just counting.

Exercise 3 — Validated Input

LEVEL 3 • INTERMEDIATE

Read an integer that must be between 1 and 100 (inclusive). If the value the user enters is outside this range, ask them to enter it again — keep asking until a valid value is provided. Then display the value that was accepted.

Hint: this is exactly the kind of problem do...while is built for: you always need to read a value at least once before you can check whether it's valid.

Exercise 4 — Digital Root

LEVEL 4 • CHALLENGE

Read a positive integer n . The “digital root” of a number is found by repeatedly summing its digits until only a single digit remains. For example, $9875 \rightarrow 9+8+7+5 = 29 \rightarrow 2+9 = 11 \rightarrow 1+1 = 2$, so the digital root of 9875 is 2.

- Use one do...while structure to sum the digits of the current number.
- Use a second, outer do...while structure to repeat this process for as long as the sum still has more than one digit.
- Display each intermediate sum, as well as the final digital root.

Hint: this requires nesting one do...while structure inside another — the inner loop computes a digit sum, and the outer loop decides whether to run it again.